

May 2026

PSHMP Core

Proactive Self-Healing Overlay Network Core

Turning any IP network into a resilient, self-healing mesh

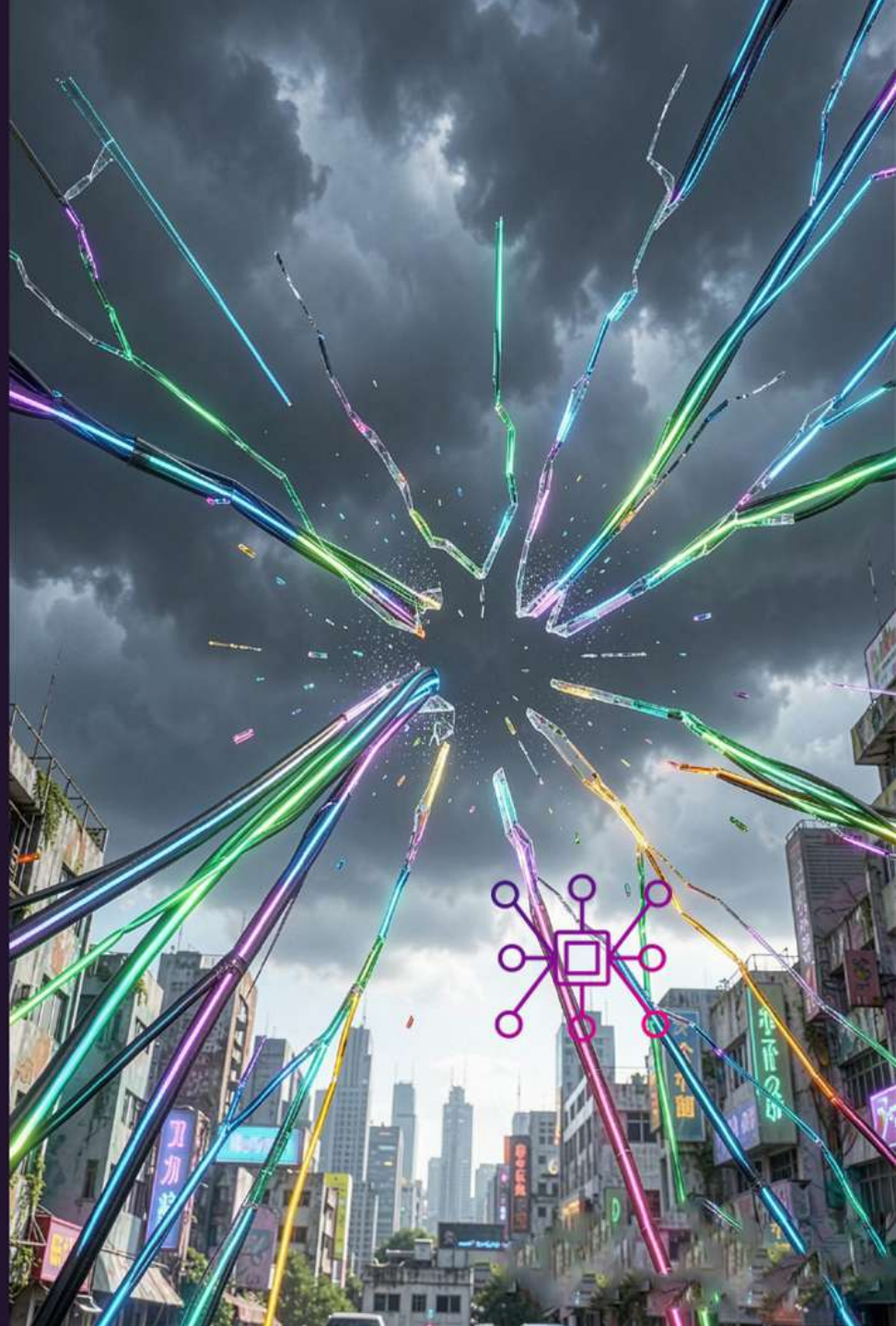
Alexander Kolomytsev, Independent Developer (Russia)

The Challenge

Bottom line: Users experience buffering.
Infrastructure burns bandwidth.

Modern distributed networks face serious reliability issues:

- **Packet loss:** 10–25% in real-world conditions
- **Recovery time:** 2–12+ seconds after node failure (BGP/OSPF convergence)
- **Service overhead:** Up to 75% of traffic is ACK-related
- **High vulnerability** to churn, DDoS, and unstable links



Our Solution

OUR SOLUTION



PSHMP Core is a lightweight overlay protocol written in Go.

Transforms existing IP networks into self-healing mesh infrastructure

Works on top of current equipment — no rip-and-replace

Deploys via Docker in minutes

Embeds into any Go-based stack



Core Innovation

Proactive Self-Healing Engine with K-Factor stability scoring

K-Factor:

Continuous per-node stability index (loss, jitter, RTT, battery)

Failure prediction:

Anticipates degradation before service impact

Proactive rerouting:

Heals the route before the user notices

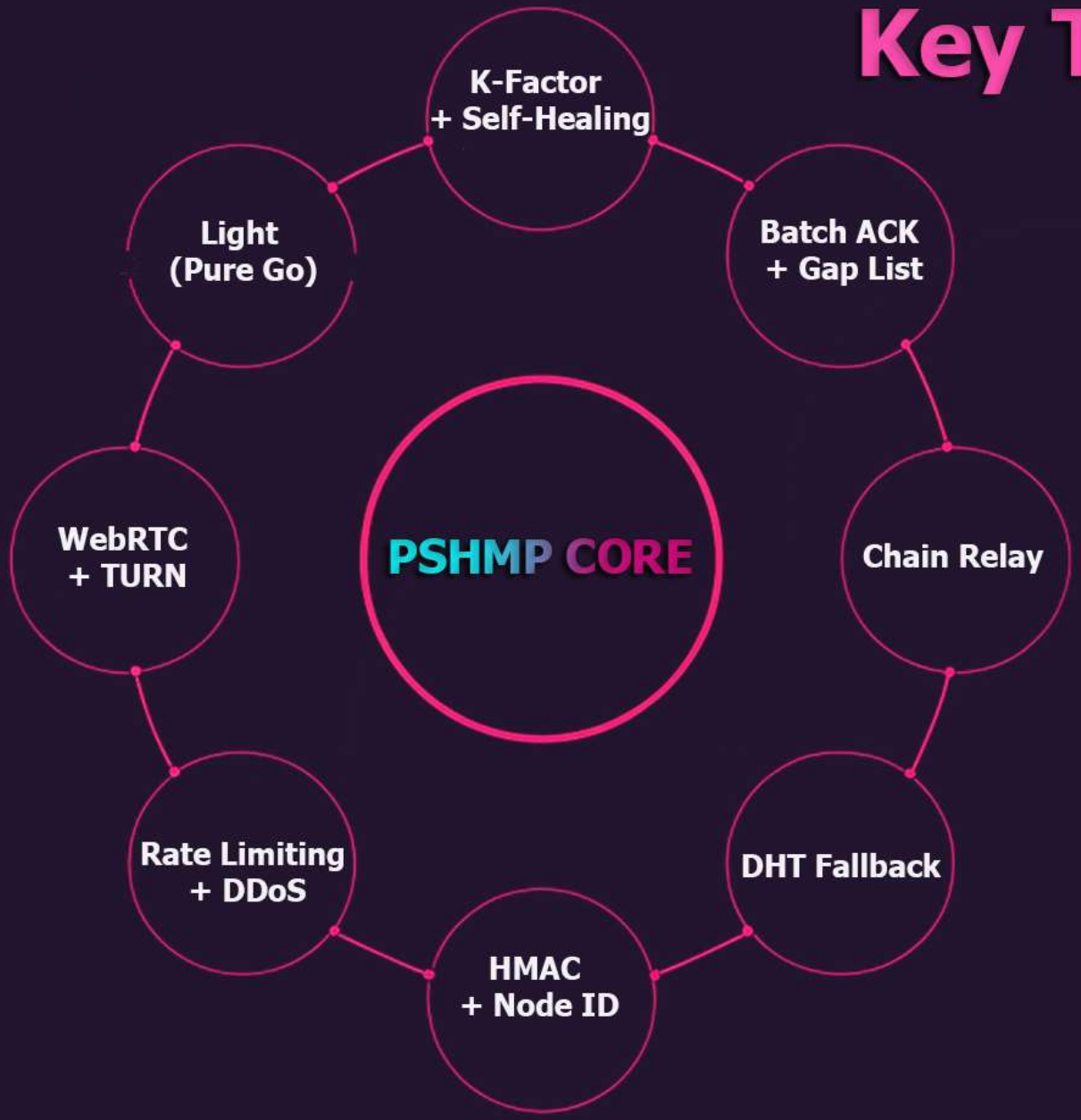
Average healing time: 380 milliseconds (measured across all test scenarios)

How It Works



Each layer operates independently yet feeds the next — creating a self-reinforcing resilience loop.

Key Technical Mechanisms



TECHNICAL ARCHITECTURE

- Batch ACK + Gap List — 75% reduction in ACK overhead
- Chain Relay — dynamic node replacement for path continuity
- DHT Fallback — decentralized peer discovery without central infra
- Rate Limiting + DDoS Protection — traffic flood mitigation
- HMAC + Unique Node ID — cryptographic integrity per hop
- WebRTC DataChannel + TURN — NAT traversal for browsers & mobile
- K-Factor + Proactive Self-Healing — 380ms recovery before failure



Performance Results (Verified by SAST Report)

Tested on Docker emulator (up to 150 nodes) + real hardware (PC, laptop, smartphone).

Demo Mode (120 nodes): 87.9% successful delivery	Stress Mode (25% packet loss + churn): 74.6% successful delivery	Real network test (3 devices): 0% packet loss (75 chunks)

Average healing speed: ~380 ms

Technology Stack

Lightweight, portable, production-grade.

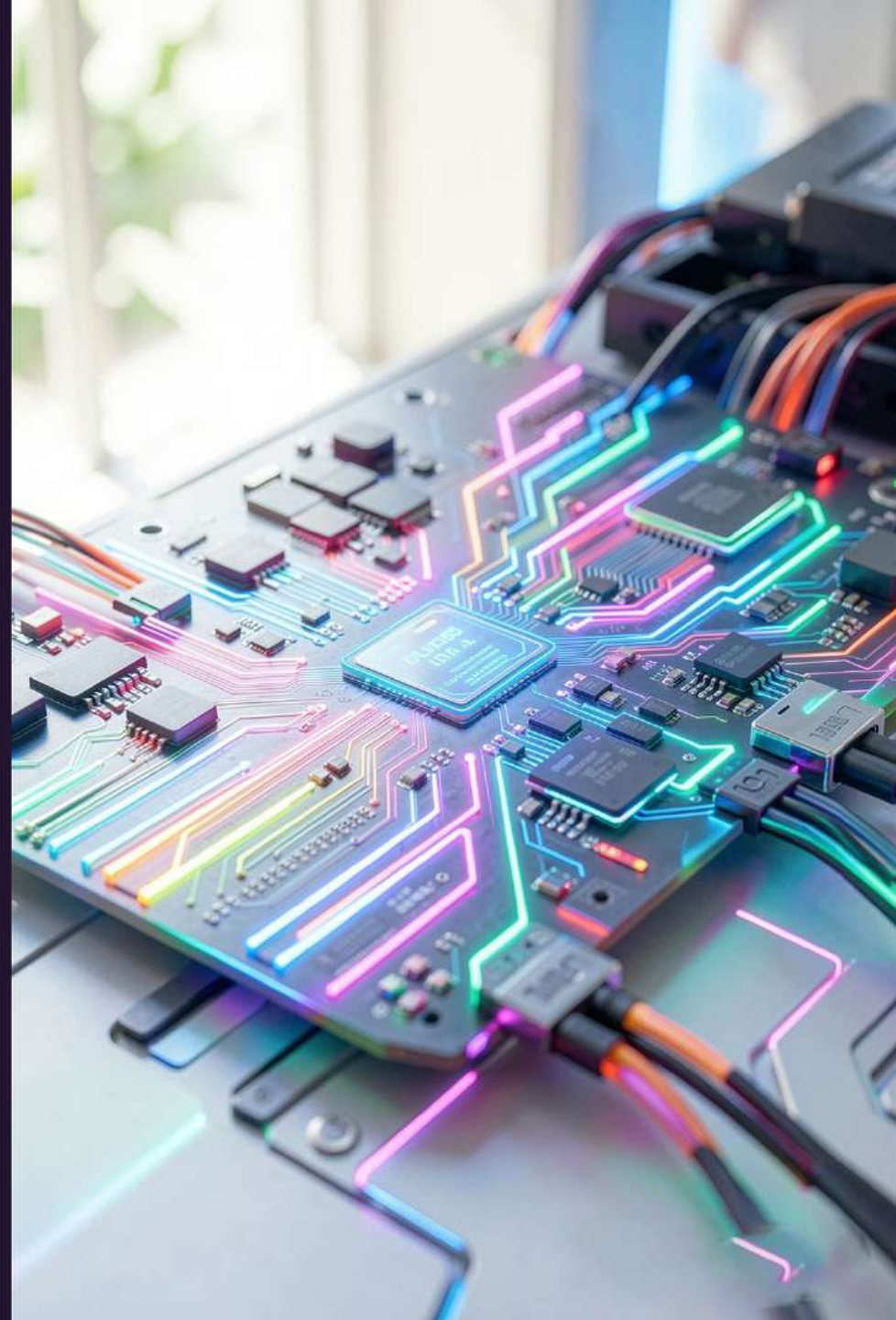
Core Language:
Go 1.26+

Transport:
UDP + WebRTC
+ TURN

Storage:
BoltDB (embedded, no external dependencies)

Deployment:
Docker
+ Docker Compose

Security:
SAST analysis passed,
HMAC + rate limiting



Use Cases



CDN and Edge
Computing optimization



5G and next-generation
mobile networks



Smart City and
Critical Infrastructure



High-resilience
environments



Hybrid Satellite
+ Terrestrial networks



Industrial IoT
and private networks

Competitive Advantages

- Pure overlay architecture (no rip-and-replace)
- Extremely fast recovery (~380 ms)
- Significant bandwidth savings (up to 75%)
- Lightweight and embeddable (Pure Go)

3 patent-pending core mechanisms



Current Status

- Pilot-ready MVP
- 10,000+ lines of clean Go code
- Docker emulator + real hardware tests completed
- Full technical documentation and security report
- Ready for pilot deployment and integration

Bonus: v3.0 scaling architecture (100k+ nodes) — available as a bonus document.



Target Markets

- Tier-1 Telecom Operators
- Global CDN Providers
- Satellite Constellations
- Enterprise & Industrial Networks
- Smart City Projects Worldwide
- Government & Emergency Services



Business Proposal

Flexible cooperation models:

- Technology licensing and source code access
- Exclusive rights
- Joint pilot project with further scaling
- Ready for NDA and technical deep-dive



Contact & Next Steps

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Sign NDA
Access code and documentation
Pilot on your infrastructure
Licensing or exclusive rights deal
Open to international cooperation and pilot projects.

Thank You.

